1. Identity

Document	STAR-PKG-STAR-LCC-EN · v1.3 · 2026-10-16
Package	star-lcc
Version	0.1.0
Capability	star.geodesy.lambert-conic
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_lcc

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_lcc-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_lcc-0.1.0-py3-none-any.whl	fd2a0ec81d72480557e9222ceee4a09831eb8d8cccd77d4b1ada5467528a8b73
star_lcc.py	b6835973a489dbb88814283e40ed14043c356d8738e24491e03540880ce181e1
star_lcc-0.1.0-source.zip	ca36a01a91d6bda8a25321892efd6630317f870c2efb8fbfce6bf5913e23deb9

5. What it does, and its limits

Lambert conformal conic projection, forward, inverse and scale factor, for cones in either hemisphere and the tangent cone; within 8e-7 m per Earth radius of PROJ and 7e-6 m of PyGeodesy on 600 cases; reproduces Snyder's published example and refuses degenerate cones.

No false easting or northing, no scale reduction k0 (the 1SP variant with k0 ≠ 1 is not offered), no named national grids. PyGeodesy was compared on 553 of the 600 cases: on the 47 whose longitude and central meridian lie on opposite sides of the date line its result differs by up to 4.7e7 m while PROJ and this module agree; that is recorded as a candidate observation about a third-party library, not verified further and not reported. For a tangent cone PyGeodesy takes the origin latitude as the parallel, so those cases were generated with lat0 = lat1.

- `lcc_forward` and `lcc_inverse`: Snyder's formulas (USGS PP 1395, 15-1 ... 15-11) with the isometric latitude written without cancellation, cones in either hemisphere, equal parallels for the tangent cone, the central meridian anywhere including across the date line. Reproduce Snyder's numerical example to the printed 0.1 m.
- `lcc_scale`: the point scale factor, exactly 1 on the standard parallels. Reproduces Snyder's k = 0.9970171.
- Measured on 600 cones and points on WGS-84 (60 tangent cones): forward within 8e-7 m per Earth radius of plane distance of PROJ and 7e-6 m of PyGeodesy; inverse within 7e-7 m (PROJ's plane coordinates) and 6e-6 m (PyGeodesy's) of the starting point; scale within 1.2e-9 of PROJ (which differentiates numerically).
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, latitudes beyond ±89, longitudes beyond ±180, standard parallels on opposite sides of the equator or within 1e-3 deg of it, an ellipsoid outside the stated ranges, a point at or beyond the apex of the cone.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_lcc_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-lcc}: Lambert conformal conic projection},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_lcc/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts